

Python

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Python

- Python is an interpreted, general-purpose, object-oriented language that is widely used nowadays for data analysis and machine learning applications.
- Syntax and indentation (tabs) are important to define your code

comment

""" multiline comments """

Tools

- VSCode
 - <https://code.visualstudio.com/download>
- Python ≥ 3.7
 - [W10-W11] Microsoft store
 - [Linux] packet manager (apt, snap, ...)
 - [MacOS] brew
 - [all] <https://www.python.org/downloads/>
- Google Colab
 - <https://colab.google/>

Python – locally

- Create folder big data
- Terminal
 - Test via terminal per python
 - Py windows installed via url
 - Python windows from store
 - Python/Python3 mac

- <https://docs.python.org/3/library/venv.html>

- Create venv

```
python3 -m venv /Users/irene/big-data-env
```

- Activate venv

```
source /Users/irene/big-data-env/bin/activate  
<venv>\Scripts\activate.bat
```

- Open folder big data in vs code
- Install library numpy
- Test w/ test.py and test.ipynb

Python

How execute the code?

- Terminal
- digit python (or python3)
- Read → Eval → Print → Loop
- Each expression is read, evaluated and the output is directly printed

```
$ python
Python 3.12.0 (main, ...)
>>> 2 + 3
5
>>> name = 'Alice'
>>> print(f'Hello, {name}!')
Hello, Alice!
>>> exit() # or Ctrl+D /
Ctrl+Z
```

Python

How execute the code?

- Script from terminal

python [file_name.py]

hello.py

```
import sys

if __name__ == '__main__':
    name = sys.argv[1]
    if len(sys.argv) > 1 else 'world'
    print(f'Hello {name}!')
```

```
$ python hello.py
Hello world!
$ python hello.py Alice
Hello Alice!
```

Python

How execute the code?

- Notebooks
- File divided into blocks executed separately

```
hello.ipynb
```

```
print(f'Hello world!')
```

```
>>> Hello world!
```

```
name = 'Alice'
```

```
print(f'Hello, {name}!')
```

```
>>> Hello, Alice!
```

Data types

- Integers
 - Represented with 32-bits, can be converted from others formats with `int()`
- Reals
 - Can be obtained via built-in function `float()` with range $[-10^{308}, 10^{308}]$ a number can be printed as -3.456 or in scientific notation -3.456e-7
- Booleans
 - Related to logical operations and to relational operator and can have only True or False values

Data types

Type	Examples	Note
int	42, -7, 0xFF, 0b1010	arbitrary precision
float	3.14, 1e-3, float('inf')	IEEE 754 double (64 bit)
complex	2+3j, complex(1,-1)	real part + imaginary part
bool	True, False	subclass of int (1/0)
bytes	b'abc', bytes(4)	immutable byte sequence
NoneType	None	No value

Data types – examples

```
n = 42
```

```
pi = 3.14159
```

```
ok = True
```

```
type(n) # <class 'int'>
```

```
isinstance(n, int) # True
```

Data types – examples

- To output/display a python object use `print()` function
- To print well-formatted strings, use `print(f'')`
 - `pi = 3.14159265358979`
 - `print(f'Pi is {pi:1.5f}')` -> `Pi is 3.14159`

Data types – strings

- Strings can be obtained via `str()` or within single or double apex notation
- Escape characters are allowed
`\n, \t, \b, \\, \', \"`
- Can be concatenated via `+` operator
- Cannot be modified, if manipulation is needed a new object is built.
- A sequential access can be performed with a direct positional indexing from zero.

Data types – strings

- `s.lower()` and `s.upper()` return string `s` in lowr/upped case
- `s.strip()` removes any white spaces at the beginning and at the end of the string
- `s.isalpha()`, `s.isdigit()`, `s.isspace()` returns `True` if ALL chatacters in a string are alphanumeric, numeric, spaces
- `s.startswith('other')` and `s.endswith('other')` check if string `s` starts/ends with string `other`

Data types – strings

- `s.find('other')` search other in s and returns the starting index of the first occurrence, otherwise -1
- `s.replace('old', 'new')` returns a string where all the occurrences of old are substituted with new
- `s.split(delim')` return a list of sub-strings divided by the given delimiter. Default delimiter is a single space.
- `s.join(list)` concatenates elements of a given list in a string with s as delimiter.

Data types – lists

- List can be defined as `a = [element, element, ...]`
- Can contain different data types of elements simultaneously
- Can be accessed via array notation and slicing
- Can be modified with in-place operations

Data types – lists

- `len(a)` return length of a given list
- `a.pop(0)` removes first element
- `a.append('Monica')` appends given element at the end of a list
- `a.insert(1, 'Joey')` insert given element at given position
- `list.extend(['Rachel', 'Phoebe'])` extends the given list with another
- `print(list.index('Joey'))` returns index of a given element
- `list.remove('Joey')` if the element is found, is removed from a list
- `print(list[1:-1])` prints list from index 1 to second-to-last element

Data types – tuple

- Tuple can be defined as $a = (\text{element}, \text{element}, \dots)$
- Can contain different data types of elements simultaneously
- Are ordered and immutable

Data types – sets

- Sets can be defined as $a = \{\text{element1}, \text{element2}, \dots\}$
- Can contain different data types of elements simultaneously
- Automatic deduplication
- Support native algebraic operations as union, intersection, difference

Data types – dictionaries

- Dictionaries can be defined as $d = \{k1: v1, k2: v2\}$
- Are a key-value hash tables
- Keys must be immutable types, no restrictions for values types

Data types

Type	Examples	Note
str	<code>'bye', "hello", r'\n'</code>	immutable, ordered
list	<code>[1, 'a', 3.0]</code>	ordered, allows duplicates
tuple	<code>(1, 'a', 3.0)</code> or <code>1</code>	immutable, ordered
set	<code>{1, 2, 3}</code> or <code>set()</code>	unordered, no duplicates
dict	<code>{'k': v, 1: [2, 3]}</code>	key-value pairs, ordered

Data types – examples

```
msg = 'Python 3' # string

lst = [10, 20, 30, 10] # list

tpl = (10, 20, 30) # tuple

st = {10, 20, 30} # set

d = {'name': 'Anna', 'age': 25}
d['name'] # value access
```

Data types – conversions

```
int('42')           # 42
float('3.14')       # 3.14
str(100)            # '100'
bool(0)             # False (0, None, [] → False)
list('abc')         # ['a', 'b', 'c']
tuple([1,2,3])      # (1, 2, 3)
set([1,1,2,3])      # {1, 2, 3}
dict(a=1, b=2)      # {'a': 1, 'b': 2}
```

Data types – slicing and indexing

- Strings, list and tuples can be accessed with the following notation `[start:stop:step]`

- Start -> inclusive first index
- Stop -> exclusive upper limit
- Step -> stride increment size

Hello

0	1	2	3	4
-5	-4	-3	-2	-1

- Their length can be obtained via `len()` function
- Note that stop position is excluded (you arrive at `len-1`)
- You can use a variant of the standard indexing

Data types – slicing and indexing

```
a = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```
a[1:4]
```

```
>>> [1, 2, 3]
```

```
a[1:6:2]
```

```
>>> [1, 3, 5]
```

```
a[::-1]
```

```
>>> [9, 8, 7, 6, 5, 4, 3, 2, 1, 0]
```

Operators

Python Operators Precedence

Operator	Description
**	Exponentiation (raise to the power)
~ + -	Ccomplement, unary plus and minus (method names for the last two are +@ and -@)
* / % //	Multiply, divide, modulo and floor division
+ -	Addition and subtraction
>> <<	Right and left bitwise shift
&	Bitwise 'AND'
^	Bitwise exclusive 'OR' and regular 'OR'
<= < > >=	Comparison operators
<> == !=	Equality operators
= %= /= //= -= +=	Assignment operators
*= **=	
is is not	Identity operators
in not in	Membership operators
not or and	Logical operators

Sorting

- Sorting of an iterable object can be obtained using the following function

```
sorted(iterable, key=None, reverse=False)
```

- It return a list of ordered element in ascending order.
 - `key` refers to a function that works as ordering criteria
 - `reverse=True` enables descending ordering.
- Lists have the built-in method `sort()` for in place ascending sorting and returns `None`

Control functions – if elif else

```
if condition:
    instruction
elif condition:
    instruction
...
else:
    instruction
```

```
x = 7
if x < 0:
    print('negative')
elif x == 0:
    print('zero')
else:
    print('positive')

# ternary condition
sign = 'pos' if x > 0 else 'non-pos'
```

Control functions – for

```
for iter in iterable:  
    instruction
```

```
for i in range(5): # 0,1,2,3,4  
    print(i)  
  
# iterates over iterable objects  
# string, tuple, sets  
for fruit in ['apple', 'pear', 'kiwi']:  
    print(fruit)
```

Control functions – for

for iter **in** iterable:
instruction

```
# enumerate: index + value
for i, v in enumerate(['a', 'b', 'c']):
    print(i, v) # 0 a, 1 b, 2 c

# items: for dictionary
dd = dict(z=-1, y=-2)
for k, v in dd.items():
    print(k, v) # z -1, y -2
```

Control functions – for

```
for iter in iterable:  
    instruction
```

```
# break / continue / else  
for n in range(10):  
    if n == 5:  
        break  
    if n % 2:  
        continue  
    else:  
        print('completed')
```

Control functions – List Comprehension

The python-way to create dynamically a list

```
newlist = [expression for item in iterable]
```

```
# classic way
fruits = ["banana", "cherry", "kiwi", "lemom", "mango"]
newlist = []
for x in fruits:
    newlist.append('*'+x)
print(newlist)

# list comprehension
newlist_2 = ['*'+x for x in fruits]
print(newlist_2)
```

Control functions – List Comprehension

```
newlist = [expression for item in iterable if  
            condition == True]
```

```
# classic way  
newlist = []  
#enumerate outputs the index and the value  
for idx, x in enumerate(fruits):  
    if idx%2==0:  
        newlist.append(x)  
print(newlist)  
  
# list comprehension  
newlist_2 = [x for idx, x in enumerate(fruits) if idx%2==0]  
print(newlist_2)
```

Control functions – Dictionary Comprehension

The python-way to create dynamically a list

```
newlist = [expression for item in iterable if  
            condition == True]
```

```
dict1 = {'a': 1, 'b': 2, 'c': 3, 'd': 4, 'e': 5}  
  
dict2 = {}  
for k, v in dict1.items():  
    dict2[k] = v**2  
print(dict2)  
  
# dict comprehension  
dict3 = {k:v**2 for k,v in dict1.items()}  
print(dict3)
```

Control functions – while

while condition:
instruction

```
n = 1
while n < 100:
    n *= 2
    print(n) # 128
```

Control functions – match/case

match variable:

```
case value_1:  
    instruction  
  
case value_2:  
    instruction  
  
case _:  
    instruction
```

```
command = 'quit'  
match command:  
    case 'quit' | 'exit':  
        print('exit program...')  
    case 'help':  
        print('Help mode...')  
    case _:  
        print('Unknown command')
```

Python 3.10+

Control functions – try/except/finally

```
try:
    except erroType:
        instruction
    else:
        instruction
    finally:
        instruction
```

```
try:
    x = int(input('Number: '))
    print(10 / x)
except ValueError:
    print('Invalid input')
except ZeroDivisionError:
    print('Cannot divide per zero')
except Exception as e:
    print(f'Error: {e}')
else:
    print('No errors')
finally:
    print('Always print')
```

Functions

```
def say_hello(name, times):  
    """say hello multiple times."""  
    for _ in range(times):  
        print(f'Hello, {name}!')  
  
say_hello('Anna', 2) # order counts  
say_hello(times=3, name='Bob') # keyword-based
```

Functions – default and optional values

```
def power(base, esp=2, *, verbose=False):  
    # parameters after * are only keyword  
    result = base ** esp  
    if verbose:  
        print(f'{base}^{esp} = {result}')  
    return result
```

```
power(3) # 9
```

```
power(2, 10) # 1024
```

```
power(2, verbose=True) # 2^2 = 4
```

Functions – lambda functions

```
square = lambda x: x ** 2
```

```
square(5) # 25
```

```
# useful with sorted, map, filter
```

```
data = [('Bob', 30), ('Anna', 25), ('Luca', 35)]
```

```
sorted(data, key=lambda t: t[1]) # age sorting
```

Functions – *args and **kwargs

```
def num_suo(*numbers): # args: tuple
    return sum(numbers)
```

```
num_suo(1, 2, 3) # 6
```

```
num_suo(1, 2, 3, 4) # 10
```

```
def show_f(**info): # kwargs: dictionary
    for k, v in info.items():
        print(f'{k}: {v}')
```

```
show_f(name='Anna', eta=25, citta='PA')
```

Functions – unpacking

```
def vectors (x, y, z) :  
    return x + y + z
```

```
coords = [1, 2, 3]
```

```
vectors (*coords) # unpacking list
```

```
params = {'x':1, 'y':2, 'z':3}
```

```
vectors (**params) # unpacking dictionary
```

Functions – parameter order

Parameter order in Python Function Signature

```
def f(pos1, pos2, *args, kw1='x', **kwargs) :  
    pass
```

*args -> positional values, as tuple

*kwargs -> key-value pairs, as dictionary

positionals values / default and/or optional arguments / * args /
keyword-only / **kwargs

Class

- Each item can be modelled as an object with attributes and methods (object's functions)

```
class Person:
    def __init__(self, name, age):
        self.name = name # attribute
        self.age = age

    def greet(self):
        return f"Hi! I am {self.name}."

    def __str__(self):
        return f"Person({self.name}, {self.age})"

user = Person("Alice", 30)
print(user.greet()) # Hi! I am Alice.
print(user) # Person(Alice, 30)
```

Import

- Import of the entire module or library

```
import math  
math.sqrt(16) # namespace access
```

- Import w/ alias

```
import numpy as np  
np.array([1,2,3])
```

Import

- Import of specific names

```
from os.path import join, exists  
join('/home', 'user') # w/o prefix
```

- Import all (bad, may overridden names)

```
from math import *  
sqrt(4)
```

Input/Output operations

- Keyword *with* delimits a code block inside which the file is opened and can be both written or read, depending the opening mode

```
with open("data.txt", mode="r", encoding="utf-8") as f:  
    content = f.read()  
  
# "r" = read | "w" = overwrite | "a" = append | "r+" =  
read/write
```

Input/Output operations

- Writing

```
with open("log.txt", "w", encoding="utf-8") as f:  
    f.write("Line 1: Server OK\n")
```

- Reading

```
with open("log.txt", "r", encoding="utf-8") as f:  
    content = f.readlines() # reading line by line  
  
for line in content:  
    print(line)
```

Input/Output operations – JSON

JSON (JavaScript Object Notation) is a dictionary-like data format

```
import json

data = {"user": "Alice", "role": "Admin"}

# Save dict as JSON file
with open("user.json", "w") as f:
    json.dump(data, f, indent=2)

# Read JSON file into dict
with open("user.json", "r") as f:
    loaded = json.load(f)
```

Input/Output operations – CSV

CSV (Comma Separated Values) is a tab-like data format

```
import csv

# Writing dictionary rows to CSV
with open("items.csv", "w", newline="") as f:
    writer = csv.writer(f)
    writer.writerow(["id", "name", "price"])
    writer.writerow([101, "Book", 19.99])

# Reading rows back
with open("items.csv", "r") as f:
    reader = csv.DictReader(f)
    for row in reader:
        print(row["name"])
```

Python

- Python 3 docs: <https://docs.python.org/3/>
- Tutorial: <https://docs.python.org/3/tutorial/>
- Built-in Functions:
<https://docs.python.org/3/library/functions.html>
- The Python Standard Library:
<https://docs.python.org/3/library/index.html>

Python – Useful resources

- Google's Python Class: <https://developers.google.com/edu/python>
- Google's Python Class exercises: <https://developers.google.com/edu/python/google-python-exercises.zip>
- W3Schools Python: <https://www.w3schools.com/python/>

Hands on exercises!

E1. Statistical measures

- Calculate mean, variance and standard deviations by formulas and double check with native functions in statistics module

E2. Degrees to Radians

- Implement a radians function that returns the radian equivalent of a degree. Use the following formula:

$$\text{rad} = \text{degrees} * \pi / 180$$

- Use this function to print a chart showing the radian equivalent of all degrees ranging from 1° to 180°. Use two digits of precision for the results. Print the outputs in a neat tabular format.

E3. Duplicate Elimination

- In organizations, a list of email addresses is often compiled for marketing purposes. However, duplicate email addresses need to be removed from this list. Write a function that receives a list and returns a list containing only unique values. Test your function with a list of email addresses.

E4. Counting Votes

- Write a script that uses a dictionary to determine the number of votes received by candidates in an election. The votes are concatenated in a string where each vote is separated from the next by a comma.
- [Hint: split can take in an argument for the specific delimiter you wish to use in a string.]

E5. Class Average – TXT

- Write code that enables you to store any number of grades into a grades.txt plain text file.
- Write code that reads the grades from the grades.txt file you created in the previous exercise.
- Display the individual grades and their total, count and average.

E6. Class Average – CSV

- An instructor teaches a class in which each student takes three exams. The instructor would like to store this information in a file named `grades.csv`. Write code that enables an instructor to enter each student's first name and last name as strings and the student's three exam grades as integers.
- Use the `csv` module to write each record into the `grades.csv` file. Each record should be a single line of text in the following CSV format:

`firstname, lastname, exam1grade, exam2grade, exam3grade`

- Read the `grades.csv` and display the data in tabular format.

E7. Class Average – JSON

- Use the json module to write the student information to the file in JSON format, create a dictionary of student data in the following format:

```
gradebook_dict = {'students': [student1dictionary,  
                                student2dictionary, ...]}
```

- Each dictionary in the list represents one student and contains the keys 'first_name', 'last_name', 'exam1', 'exam2' and 'exam3', which map to the values representing each student's first name (string), last name (string) and three exam scores (integers).
- Output the gradebook_dict in JSON format to the file grades.json.
- Read the grades.json and display the data in tabular format.